LAB 3 SUBMISSION

Component Modelling & Architectural Pattern Selection

Problem Statement 51 – Virtual Technical Conference Platform

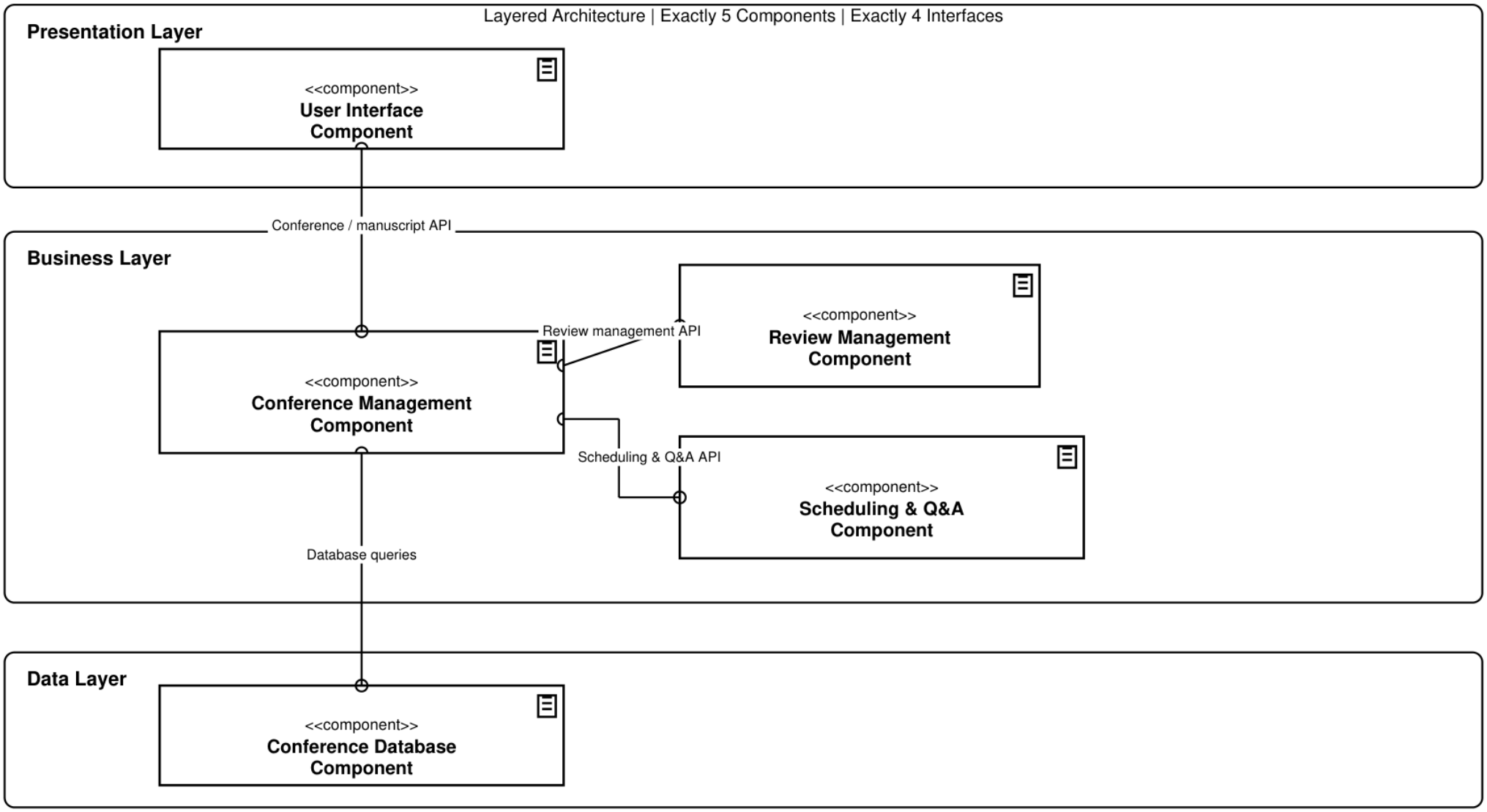
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Selected Architecture: Layered Architecture

Component Diagram + Architectural Justification

Lab 3: UML Component Diagram

Problem Statement 51 - Virtual Technical Conference Platform



Interface notation ○ Provided interface (ball) (Required interface (socket)

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Lab 3 – Architectural Justification

Problem Statement 51 – Virtual Technical Conference Platform

Architecture Selection

We chose Layered Architecture for the Virtual Technical Conference Platform.

Reason 1

The platform has clearly separable responsibilities: a user-facing layer for Conference Authors and Track Chairs, a business layer for manuscript review, conflict-of-interest handling, scheduling and live Q&A;, and a data layer for persistent conference information. This separation improves maintainability and keeps presentation logic independent from core conference operations.

Reason 2

The platform requires reliable processing of several related workflows, including manuscript anonymization, conflict-free reviewer assignment, multi-track scheduling and live Q&A.; A layered structure gives these operations a clear business-logic boundary and prevents the user interface from directly accessing stored conference data.

Security Advantage

Security controls can be concentrated in the business layer so that the presentation layer cannot directly access the database. This supports controlled access to manuscripts and review information and helps enforce the double-blind requirement by keeping author identity handling within protected application logic.

Performance Benefit

The architecture keeps the main application operations within a single structured system, avoiding unnecessary distributed-service network calls. This reduces communication overhead for operations such as schedule generation and review management, supporting the requirement that scheduling complete within 10 seconds under simulated peak load.

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